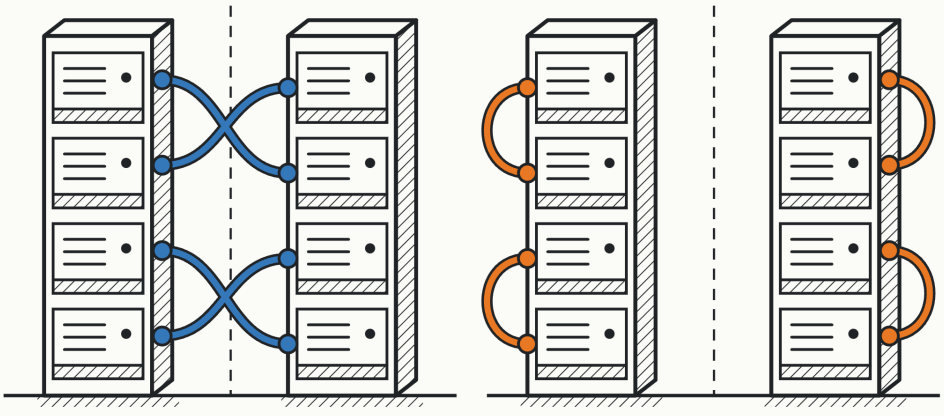
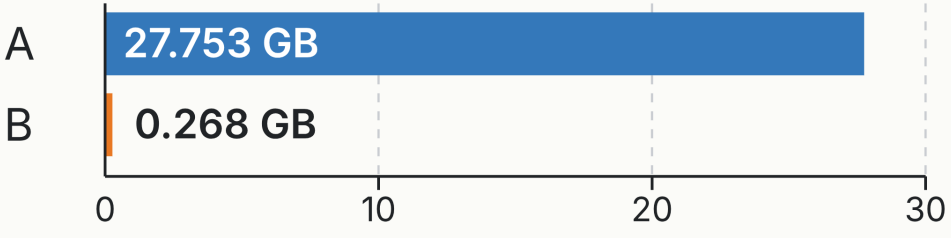
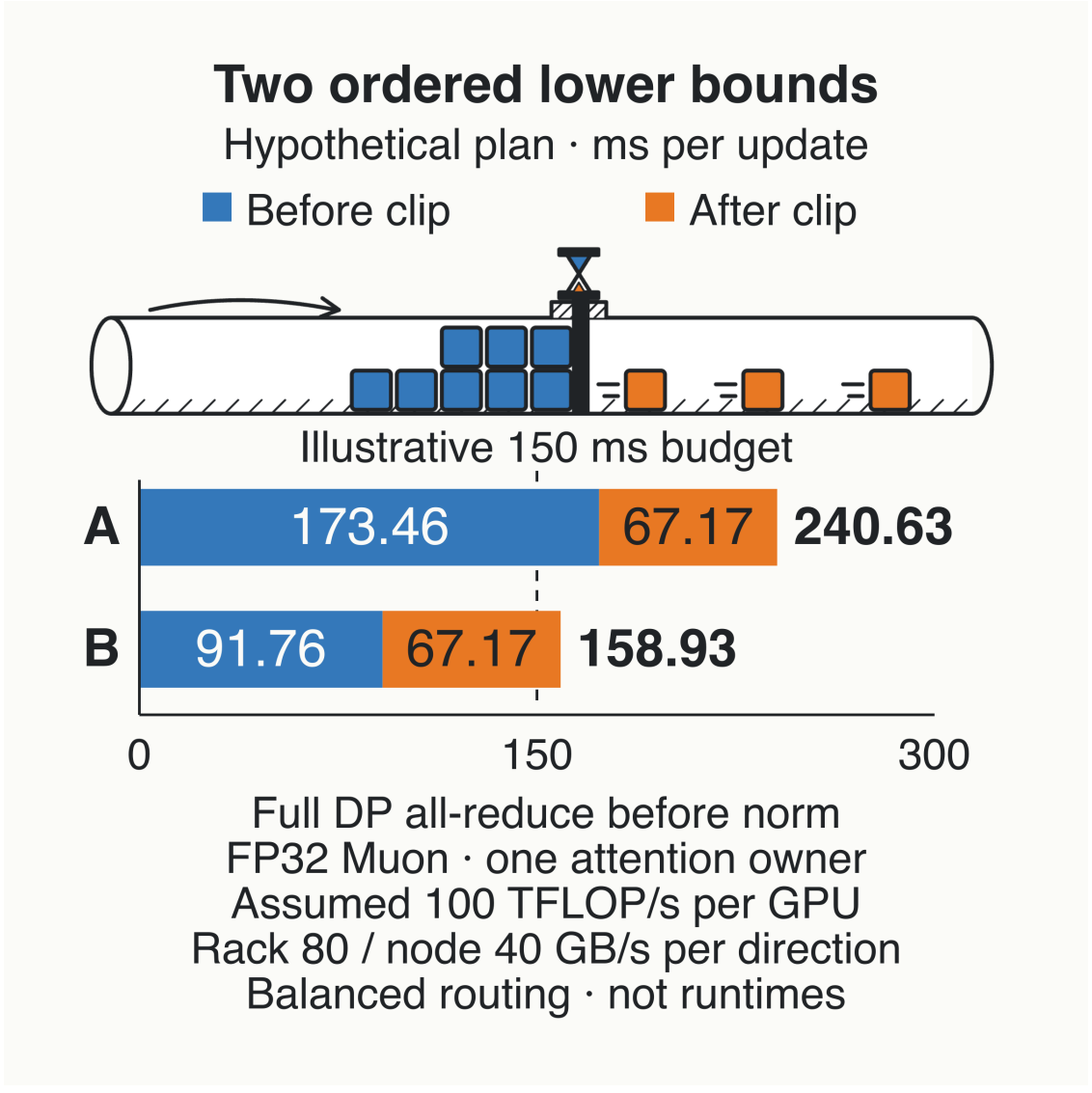
Placement moves rack traffic

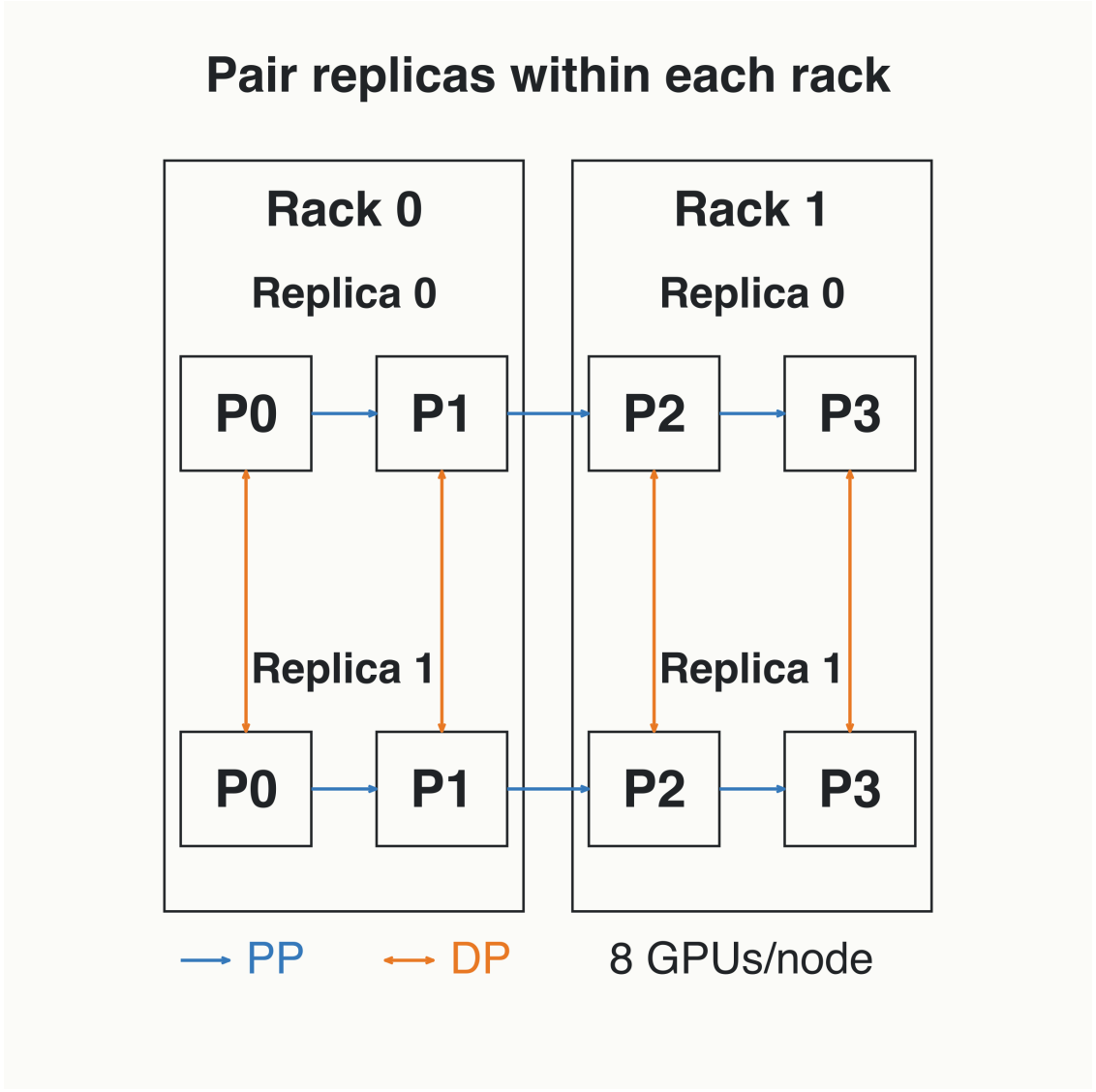


Hypothetical cross-rack GB / update
Both directions



Within rack: 0.805 → 28.290 GB
Total off-node payload unchanged


Two ordered lower bounds
Hypothetical plan · ms per update
Before clip
After clip
Illustrative 150 ms budget
A
173.46
67.17
240.63
B
91.76
67.17
158.93
0
150
300
Full DP all-reduce before norm
FP32 Muon · one attention owner
Assumed 100 TFLOP/s per GPU
Rack 80 / node 40 GB/s per direction
Balanced routing · not runtimes


Pair replicas within each rack
Rack 0
Replica 0
P0
P1
Replica 1
P0
P1
Rack 1
Replica 0
P2
P3
Replica 1
P2
P3
PP
DP
8 GPUs/node